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CSE 112 Computer Organization: Quiz 1
(SET A)

Total Time: 30 Minutes **Maximum: 30 Points**

Instructions:

1. **Write the SET name on the top of the first page of answer script for it to get evaluated.**
2. **Mobile phones, bags, and wristwatches of any kind are NOT Permitted in the examination hall.**
3. **Write all the assumptions clearly, if any. *Only reasonable assumptions will be considered if any ambiguity is found in the question*.**
4. **Institute's plagiarism policy will apply if any unfair means are observed.**
5. **Use of calculators is permitted.**

Q1. Imagine a virtual ISA having an instruction size of 32 bits and 64 registers. Suppose the ISA supports an address space of **512 KiloBytes** with byte-addressable memory. The number of **unique** Instructions/Operations supported by the processor is 64. **[10 Points]**

Syntax for Memory type instruction : <Opcode><Memory address> <Register address> <Filler Bits> *Syntax for Register type instruction* : <Opcode><Source Register Address> <Destination Register Address> <Filler Bits>

a. What are the minimum bits required to represent the opcode of instruction? **[1 Point]**
b. What are the minimum bits required to represent a register uniquely? **[1 Point]**
c. What are the minimum bits required to represent a memory address uniquely? **[1 Point]**
d. How many filler bits are required in memory type and register type instruction respectively? **[1 Point]**
e. For a memory type instruction, with hexadecimal value F4000884, determine the value of opcode, memory address and register address. **[3 Points]**
f. For a register type instruction, with hexadecimal value 58400000, determine the value of opcode, source register address and destination register address. **[3 Points]**

Q2. Solve the following: **[10 Points]**

(a) Find "x" if $(113)_x = 23$ **[2 Points]**
(b) $1100.0101_2 = (?)_{10}$ **[2 Points]**
(c) (Last two digits of your roll no.)$_{10}$ = $(?)_{16}$ **[1 Point]**
(d) Ideally, how many bits are required to represent 0.4_{10} in binary form? **[1 Point]**
  (i) Use 6 bits after decimal point to represent 0.4_{10} in binary **[1 Point]**
  (ii) Use 6 bits after decimal point to represent 0.2_{10} in binary **[1 Point]**
  (iii) Using part (d)-(ii), compute $(0.2_{10} + 0.2_{10})$ in binary and compare it with the value obtained in part (d)-(i). Report the error (if any). **[2 Points]**

Q3. Operand Op1 = $(1000)_2$, Operand Op2 = $(7)_{10}$ **[10 Points]**

(a) Evaluate the decimal value of Op1 by considering the given value in Op1 to be already present in, **[0.5*4 = 2 Points]**
  (i) 4-bit Unsigned binary form
  (ii) 4-bit Signed magnitude binary form
  (iii) 4-bit 1's complement binary form
  (iv) 4-bit 2's complement binary form

(b) Mention any one method to calculate the overflow in 2's complement addition. Derive its boolean expression also. **[2 + 1 = 3 Points]**

(c) Evaluate the following using 4-bit 2's complement arithmetic, assuming Op1 to be already in 2's complement representation, and report the final value in decimal: **[2*2 = 4 Points]**
  (i) Op1 + Op2
  (ii) Op1 - Op2

(d) Calculate the presence of overflow in parts (c)-(i) and (c)-(ii) using the boolean expression derived in (b). **[0.5*2 = 1 Point]**